The Congruence Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: CRITERIA, PARTS, ISOSCELES and REASON (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (CRITERIA)
- ☐ **A2** (CRITERIA)
- ☐ **A3** (PARTS)
- ☐ **A4** (PARTS)
- ☐ **A5** (ISOSCELES)
- ☐ **A6** (REASON)

Entrées

10 pts each · choose 4

- ☐ **E1** (CRITERIA)
- ☐ **E2** (CRITERIA)
- ☐ **E3** (PARTS)
- ☐ **E4** (PARTS)
- ☐ **E5** (ISOSCELES)
- ☐ **E6** (REASON)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (ISOSCELES)
- ☐ **S2** (PARTS)
- ☐ **S3** (REASON)
- ☐ **S4** (CRITERIA)

My skill check (circle each skill you have covered): CRITERIA · PARTS · ISOSCELES · REASON

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 CRITERIA ☐ My pick · 5 pts

In △ABC and △DEF, AB = DE = 5, BC = EF = 7, and CA = FD = 9. Which congruence criterion applies, and what is the congruence statement?

A2 CRITERIA ☐ My pick · 5 pts

In △PQR and △XYZ, PQ = XY = 7, ∠Q = ∠Y = 40°, and QR = YZ = 9. Which criterion proves △PQR ≅ △XYZ? Explain why the angle matters.

A3 PARTS ☐ My pick · 5 pts

△ABC ≅ △DEF with ∠A = 50° and ∠B = 60°. Find ∠F and name the side congruent to AC.

A4 PARTS ☐ My pick · 5 pts

△JKL ≅ △MNP with JK = 3x + 2 and MN = 20. Find x and JK.

A5 ISOSCELES ☐ My pick · 5 pts

△ABC is equilateral with AB = 2x + 1 and BC = x + 7. Find x and the measure of each angle.

A6 REASON ☐ My pick · 5 pts

Two triangles each have angles 40°, 60°, and 80°. A student says they must be congruent. Is the student right? Explain.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 CRITERIA ☐ My pick · 10 pts

△ABC has ∠B = 90°, AB = 5, AC = 13. △DEF has ∠E = 90°, DE = 5, DF = 13. Show the triangles are congruent, naming the criterion, then find EF.

E2 CRITERIA ☐ My pick · 10 pts

In △ABC and △DEF, ∠A = ∠D = 45°, ∠B = ∠E = 70°, and BC = EF = 8. Find ∠C and ∠F. Which criterion applies directly to the given information, and why is it not ASA?

E3 PARTS ☐ My pick · 10 pts

△ABC ≅ △DEF. AB = 2x + 3, DE = 5x − 12, BC = 4y − 1, and EF = y + 11. Find x, y, AB, and BC.

E4 PARTS ☐ My pick · 10 pts

△ABC ≅ △DEF with ∠A = (3x + 5)°, ∠D = (5x − 15)°, and ∠B = 70°. Find x, then ∠C and ∠F.

E5 ISOSCELES ☐ My pick · 10 pts

△ABC is isosceles with AB = AC. AB = 4x − 3, AC = 2x + 9, and BC = 3x. Find x and the perimeter of △ABC.

E6 REASON ☐ My pick · 10 pts

Segments AB and CD intersect at M, and M is the midpoint of both. Write a short proof (statements and reasons) that △AMC ≅ △BMD, naming the criterion.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 ISOSCELES ☐ My pick · 20 pts

△ABC has AB = AC = 13 and BC = 10. D is the midpoint of BC. (a) Prove △ABD ≅ △ACD, naming the criterion. (b) Use CPCTC to show AD ⊥ BC. (c) Find AD.

S2 PARTS ☐ My pick · 20 pts

△ABC ≅ △DEF. AB = 2x + y, DE = 17, BC = x + 2y, EF = 16, and AC = 3x + 1. (a) Find x and y. (b) Find DF. (c) Find the perimeter of △DEF.

S3 REASON ☐ My pick · 20 pts

In △ABC and △DEF, AB = DE = 8 and ∠A = ∠D = 35°. For each extra fact, say what it proves or that it is not enough, and why: (a) AC = DF (b) ∠B = ∠E (c) BC = EF (d) ∠C = ∠F

S4 CRITERIA ☐ My pick · 20 pts

In quadrilateral ABCD, ∠B = ∠D = 90° and AB = AD = 8. Diagonal AC = 17. (a) Prove △ABC ≅ △ADC, naming the criterion. (b) Use CPCTC to explain why AC bisects ∠BAD. (c) Find the perimeter of ABCD.